

values for each axis. If the received value for any axis crosses the pre-defined value, a corresponding 4-bit code is generated at pins 8 through 11. This code is encoded through IC1 and transmitted using RF transmitter module.

Accelerometer module. An accelerometer is an electromechanical device that measures acceleration of anything that it is mounted on. The accelerometer module used here is based on ADXL335 triple-axis accelerometer from Analog Devices. The sensor has a full sensing range of $\pm 3g$.

Arduino Uno board. Arduino is an Open Source electronics prototyping platform based on flexible, easy-to-use

hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino Uno is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, a power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller

on the board is programmed using Arduino programming language and Arduino development environment.

Robot. It comprises an RF receiver module, decoder HT12D (IC3), microcontroller PIC16F877A (IC4), operational amplifier LM324 (IC5), motor driver L293D (IC6), regulator 7805 (IC7) and a few discrete components.

Data transmitted from the wireless steering module is received by the RF receiver module and further decoded by IC3. Decoded data is fed to pins RC0 through RC3 of microcontroller IC4. Identical addresses can be selected for

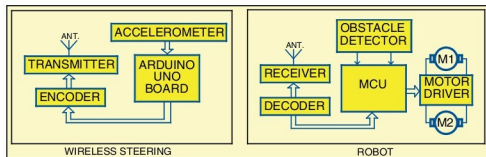


Fig. 2: Block diagram of the complete project

TABLE II
Test Points for
Wireless Steering

Test point	Details
TP0	Ground (GND)
TP1	5V
TP2	Battery voltage
TP3	Input for IC1
TP4	Input for IC1
TP5	Input for IC1
TP6	Input for IC1

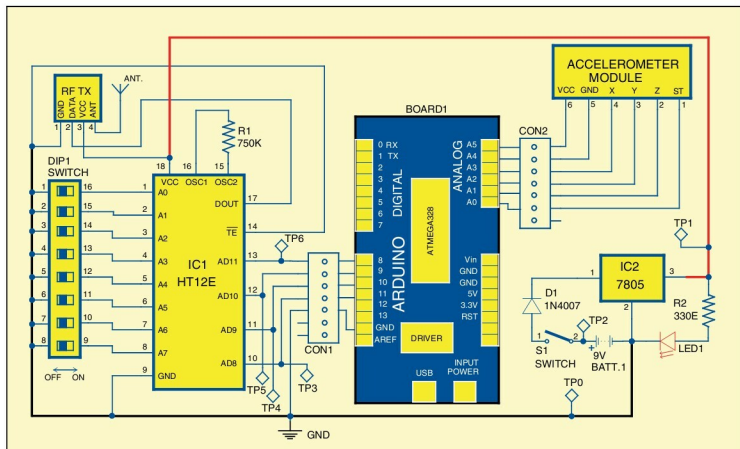


Fig. 3: Circuit of the wireless steering